

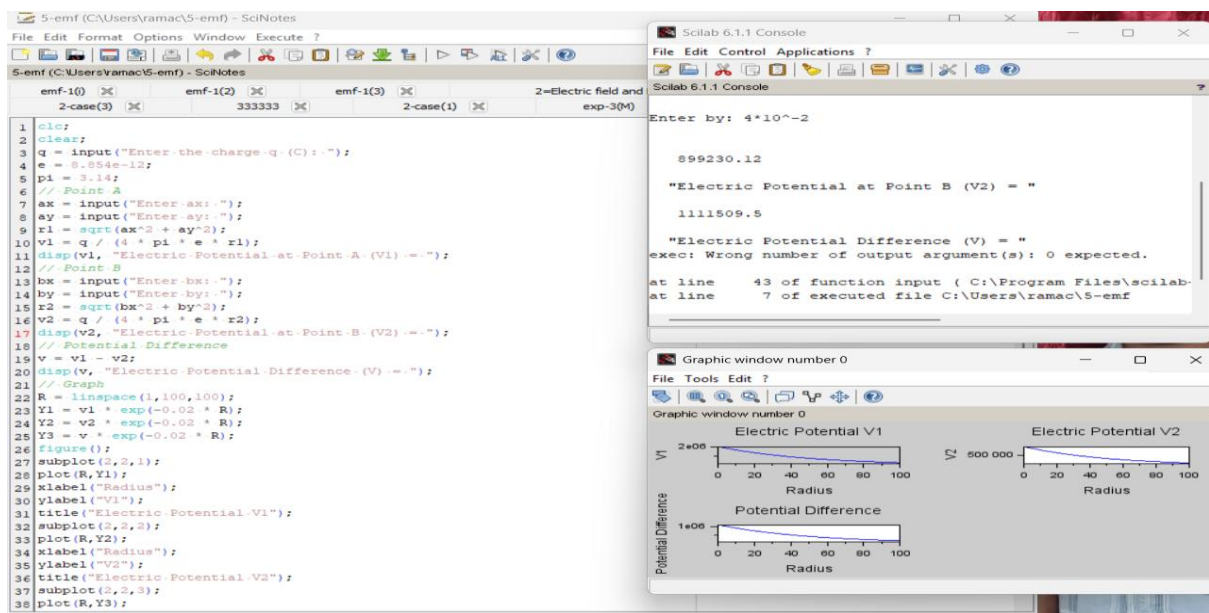
## Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

### AIM:

To write program for Electric potential difference between two points in free space medium using SCILAB.

### SOFTWARE REQUIRED: SCILAB

### PRACTICAL CALCULATION:



### THEORITICAL CALCULATIONS:

⑤  $ax = 10 \text{ cm}$      $bx = 3 \text{ cm}$   
 $ay = 2 \text{ cm}$      $by = 4 \text{ cm}$

$q = 5 \times 10^{-6}$   
 $r = 3 \text{ cm}$

①  $r_1 = \sqrt{ax^2 + ay^2}$   
 $\Rightarrow \sqrt{(0.01)^2 + (0.002)^2}$   
 $\Rightarrow \sqrt{0.0005} \Rightarrow r_1 = 0.02236 \text{ m}$

②  $v_1 = \frac{q}{4\pi\epsilon_0 r_1}$   
 $\Rightarrow \frac{5 \times 10^{-6}}{4 \times 3.14 \times 8.854 \times 10^{-12} \times 0.02236}$   
 $\Rightarrow \frac{5 \times 10^{-6}}{0.402 \times 10^{-12}} \Rightarrow 0.014 \times 10^6 \text{ V}$

③  $r_2 = \sqrt{bx^2 + by^2}$   
 $\Rightarrow \sqrt{(0.03)^2 + (0.04)^2}$   
 $\Rightarrow \sqrt{0.0009 + 0.0016} \Rightarrow \sqrt{0.0025} \Rightarrow 0.05 \text{ m}$

④  $v_2 = \frac{q}{4\pi\epsilon_0 r_2}$   
 $\Rightarrow \frac{5 \times 10^{-6}}{4 \times 3.14 \times 8.854 \times 10^{-12} \times 0.05} \Rightarrow \frac{5 \times 10^{-6}}{5.56 \times 10^{-12}} \Rightarrow 0.993 \times 10^6 \text{ V}$

\* electric potential difference  
 $V = v_1 - v_2$   
 $\Rightarrow 0.014 \times 10^6 - 0.993 \times 10^6$   
 $V = 1,115,000 \text{ V}$

